E-COMMERCE SESSION ANALYSIS

Purchase Intent Prediction & Conversion Optimization

TECHNICAL REPORT

Eva Sun

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1. Problem, Goals and Audiences

1.1 Audience & Persona

The primary audience is the business operations and marketing team of a cosmetic supply e-commerce platform serving the Russian professional beauty market.

> **Persona: E-Commerce Marketing Manager**
> Identity: Marketing or growth manager responsible for conversion rate optimisation and customer acquisition on the platform. Works cross-functionally with the product and data teams to design promotional campaigns, recommendation features, and re-engagement strategies.

Pain Points:

- No visibility into which browsing sessions are likely to convert before they end
- Unable to distinguish between casual browsers and genuine purchase-intent users in real time
- High cart abandonment rate (86.49% of cart sessions do not purchase) with no systematic re-engagement mechanism

Goals:

- Increase overall platform conversion rate beyond the current 3.43%
- Identify high-intent users before they leave the platform without purchasing
- Allocate marketing spend and platform exposure efficiently across user segments

Core Need:

To identify users most likely to make a purchase before their session ends, and to intervene with targeted product recommendations and re-engagement strategies that increase the likelihood of conversion — without incurring significant additional cost per user.

1.2 Problem Statement

The platform records approximately 4.5 million user sessions but achieves an overall purchase conversion rate of only 3.43%. Funnel analysis reveals two critical drop-off points:

- View to Cart: 81.58% of browsing sessions end without any add-to-cart action

- Cart to Purchase: 86.49% of sessions that reach the cart stage do not result in a purchase

The core business problem is: Where in the session do purchase decisions break down, and what can be done to recover high-intent sessions that currently fail to convert?

1.3 Goals

- Identify where purchase decisions break down using funnel analysis
- Predict purchase intent before a session end using a classification model
- Diagnose why high-intent sessions fail to convert
- Propose a ranking-based recommendation system to recover high-intent non-converting sessions

1.4 Criteria for Success

Criterion	Target
Funnel drop-off points identified	Both stages quantified with statistical evidence
Predictive model ROC-AUC	≥ 0.90
Session segmentation	Statistically distinct intent groups with different conversion rates
Diagnosis findings	At least 3 actionable inhibitors identified
Recommendation system	Ranking-based system validated on target session sample

1.5 Approach Overview

- Stage 1 — Data Cleaning: Event-level data cleaned and aggregated to session level
- Stage 2 — Funnel Analysis: Purchase funnel constructed to identify drop-off points
- Stage 3 — Predictive Modelling: Multiple models trained and compared; LightGBM selected as final model
- Stage 4 — Session Segmentation: Sessions segmented by predicted purchase probability
- Stage 5 — Diagnosis & Recommendation: High-intent non-converting sessions analysed; ranking-based recommendation system designed

2. Data Sources and Definitions

2.1 Dataset Overview

Attribute	Value
Source	Open CDP Project (Kaggle: mkechinov/ecommerce-events-history-in-cosmetics-shop)
Time Period	October 2019 – February 2020
Raw Records	~19.6 million event rows
Unit of Analysis	Session (one continuous platform visit)
Total Sessions (cleaned)	4,535,940
Platform Type	Professional beauty supply e-commerce (Russian market)

2.2 Data Dictionary

Field	Type	Description
event_time	datetime	Timestamp of user action, converted from UTC to Moscow Standard Time (UTC+3)
event_type	string	Type of action: view, cart, purchase, remove_from_cart
product_id	string	Unique product identifier
category_id	integer	Numerical category identifier (525 unique values)
category_code	string	Hierarchical category label; 98% missing, not used in analysis
brand	string	Product brand name; unknown filled where missing
price	float	Product price in USD; negative values indicate returns
user_id	string	Unique user identifier
user_session	string	Unique session identifier
is_return	boolean	Engineered: True where event_type == purchase and price < 0
hour	integer	Hour of day (0–23) in Moscow time
day_of_week	string	Day of week in Moscow time

2.3 Session-Level Features

Feature	Description
session_duration_min	Total session length in minutes (null if > 1440 min)
num_views / num_cart / num_purchase / num_remove	Count of each event type per session
unique_products	Number of distinct products interacted with
avg_price / max_price	Average and maximum product price in session
view_depth	unique_products / num_views, capped at 99.9th percentile (30)
single_view	Binary: 1 if session contains only one view event
hesitation_rate	num_remove / num_cart, capped at 1
converted	Binary target variable: 1 if session contains a purchase

2.4 Data Cleaning Summary

Step	Action	Rows Affected
Missing user_session	Dropped	4,598
Exact duplicates	Dropped	Pipeline ingestion errors
Negative price on non-purchase events	Dropped	5
Zero price purchases	Dropped	1
Negative price purchases (returns)	Flagged as is_return=True, retained	119
Price outliers (> 99.9th pct)	Retained — legitimate premium products	18,363
Session duration > 1440 min	Duration set to null, session retained	36,195
Timezone	Converted UTC → Moscow Standard Time (UTC+3)	All records

3. Patterns, Trends and Insights

3.1 Funnel Analysis

Stage	Sessions	% of View Sessions	Drop-off Rate
View	4,280,699	100%	—
Cart	788,430	18.42%	81.58% lost
Purchase	106,526	2.49%	86.49% of cart sessions lost

Drop-off 1 (View to Cart) represents the largest absolute loss at 3,492,269 sessions. Drop-off 2 (Cart to Purchase) is strategically more important as these users have already demonstrated purchase intent. The overall conversion rate of 3.43% across 4,535,940 sessions represents a significant opportunity for recovery.

> *Note: The funnel base (4,280,699 view sessions) is lower than the total cleaned sessions (4,535,940) because approximately 255,000 sessions contained only cart or purchase events with no recorded view event, and are therefore excluded from the funnel analysis.*

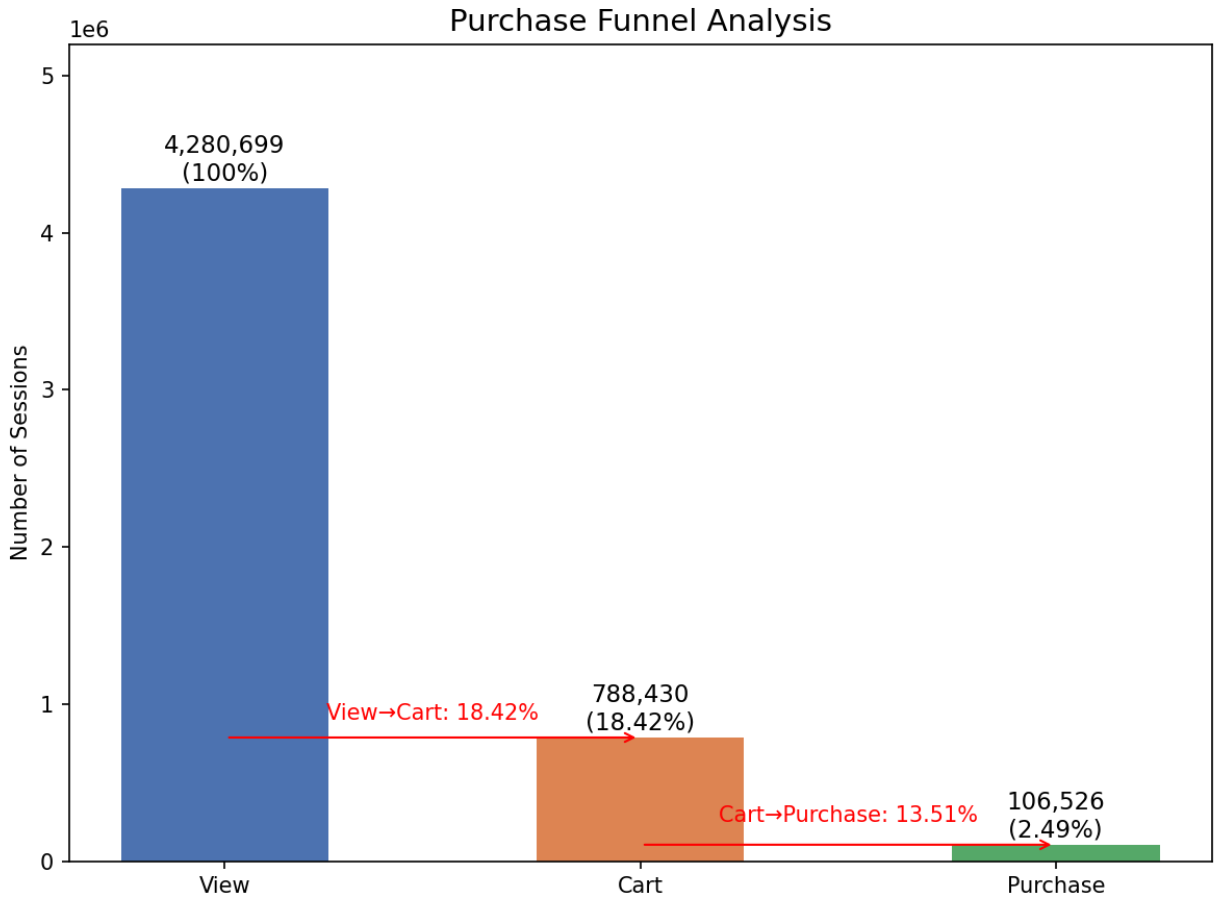

Figure 1: Purchase Funnel Analysis

3.2 Temporal Patterns

Converting sessions were disproportionately concentrated between 10:00 and 14:00 Moscow time, showing the highest conversion rates across all days of the week. An unexpected finding is that 12:00 AM (midnight) also exhibits the highest overall conversion rate at 4.54%, suggesting a segment of highly goal-oriented professional buyers active during late night hours.

In contrast, early morning hours between 1:00 and 6:00 AM consistently show the lowest conversion rates across all days. Friday demonstrates the strongest overall conversion performance, while Saturday and Sunday show consistently lower conversion rates throughout the day, indicating that weekend sessions tend toward recreational browsing rather than purposeful purchasing.

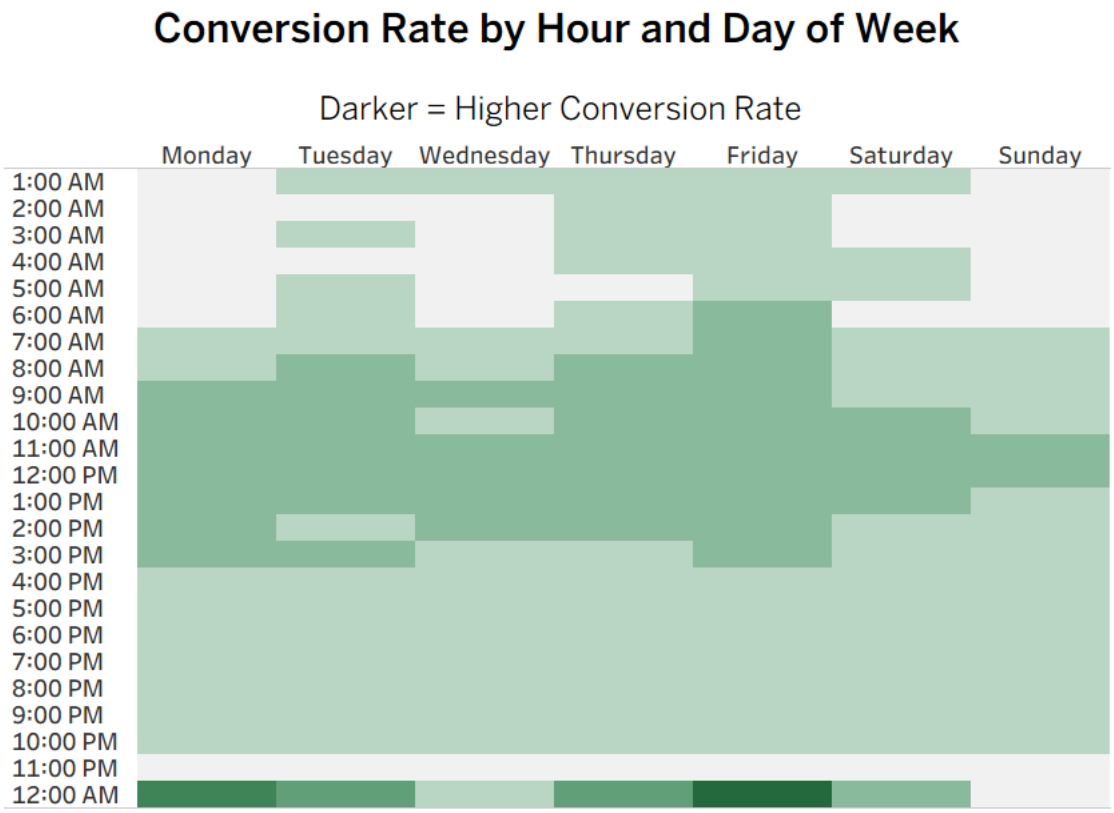

Figure 2: Hourly Conversion Heatmap by Day of Week

3.3 Key Behavioural Insights

Insight	Evidence	Implication
Engagement drives conversion	total_events 56% higher in converting sessions	Increase session depth through related product exposure

Insight	Evidence	Implication
Exploration breadth matters	unique_products 21% higher in converting sessions	Surface more product variety during sessions
Price sensitivity inhibits conversion *	avg_price 7.2% higher in non-converting sessions	Recommend cheaper alternatives in same category
Timing affects conversion	Daytime weekday sessions convert significantly better	Target re-engagement during high-conversion windows
Hesitation is not the primary barrier	hesitation_rate only -3.2% difference	Cart removal is not a key inhibitor

** Early-stage observation based on aggregate session data (all sessions). Subsequent High Intent segment analysis in Section 5.1 shows that converted users actually pay 4.1% more on average, suggesting that product comparison builds purchase confidence rather than driving price-seeking behaviour. The recommendation strategy in Section 6 therefore prioritises broader product discovery over cheaper alternatives.*

3.4 Tableau Visualisation

A Tableau workbook was developed containing nine analytical views: Data Overview, Session Duration, Event Distribution, Price Analysis, Purchase Funnel, Hourly Conversion Heatmap, Session Segmentation by Intent, Behavioural Diagnosis Comparison, and Brand Distribution. Each view provides targeted insights to support business decision-making.

4. Predictive Model

4.1 Objective and Approach

A binary classification model was developed to predict purchase probability at the session level. The target variable converted indicates whether a session resulted in at least one purchase. Given severe class imbalance (3.43% positive class, ratio 1:28), all models were trained with class_weight='balanced'. Stratified train-test splitting (80/20) preserved class ratios across both sets.

4.2 Feature Set

Feature	Rationale
num_views	Volume of browsing activity

Feature	Rationale
num_cart	Add-to-cart behavior as intent signal
num_remove	Cart removal as hesitation signal
unique_products	Breadth of product exploration
session_duration_min	Time investment in session
view_depth	Ratio of unique products to views
single_view	Indicator of minimal engagement
hesitation_rate	Proportion of cart additions removed

Note: cart_abandoned was excluded after initial testing revealed feature leakage. It was derived directly from num_purchase, which also defines the target variable converted, producing an artificially perfect ROC-AUC of 0.996.

4.3 Model Comparison

Model	ROC-AUC	Class 1 Recall	Class 1 Precision	Class 1 F1
Logistic Regression (default threshold 0.5)	0.9143	0.84	0.16	0.27
Logistic Regression (optimised threshold 0.654)	0.9143	0.70	0.23	0.34
Random Forest	0.9029	0.45	0.48	0.47
LightGBM (threshold 0.3) — FINAL	0.9764	0.99	0.21	0.34

4.4 Final Model: LightGBM

LightGBM was selected as the final model based on its superior ROC-AUC of 0.9764 and recall of 0.99. The model was trained with 500 estimators and learning rate 0.05. A threshold of 0.3 was selected as the minimum eligibility cutoff for recommendations.

The low precision (0.21) is intentional and justified by the business context. As the recommendation system operates through in-platform exposure rather than paid messaging, the marginal cost of an additional recommendation is near zero. Therefore, the trade-off shifts from monetary cost to traffic allocation efficiency, making high recall the priority over high precision.

Metric	Value	Interpretation
ROC-AUC	0.9764	In 97.64% of cases, model correctly ranks a buyer above a non-buyer
Recall (Class 1)	0.99	99% of true purchasers are identified
Predicted buyers	185,658	Sessions eligible for recommendation exposure
Actual buyers	38,904	True purchasers in test set
Threshold	0.3	Minimum probability for recommendation eligibility

4.5 Ranking-Based Recommendation Strategy

Rather than applying binary classification, the final system uses predicted purchase probabilities as a continuous ranking score. This preserves the full probability signal and optimises traffic allocation:

Tier	Threshold	Strategy
High Exposure	Top 20% by probability	Priority placement, featured recommendations
Normal Exposure	50th – 80th percentile	Standard recommendation carousel
No Exposure	Below 50th percentile	Excluded from recommendation engine

4.6 Session Segmentation

Segment	Threshold	Sessions	% of Total	Conversion Rate
High Intent	prob ≥ 0.733	453,182	10.0%	31.1%
Medium Intent	0.023 ≤ prob < 0.733	682,463	15.0%	2.1%
Low Intent	prob < 0.023	3,400,295	75.0%	0.002%

The High Intent segment conversion rate (31.1%) is 1,555x higher than Low Intent (0.002%), confirming the segmentation is statistically meaningful. Of 453,182 High Intent sessions, 312,160 (68.9%) did not convert, representing the primary target for intervention.

5. Diagnosis of Non-Converting High-Intent Sessions

The diagnosis compares 141,022 High Intent converted sessions against 312,160 High Intent non-converting sessions, as defined by the final LightGBM model (threshold 0.733 for High Intent segment).

5.1 Behavioural Comparison

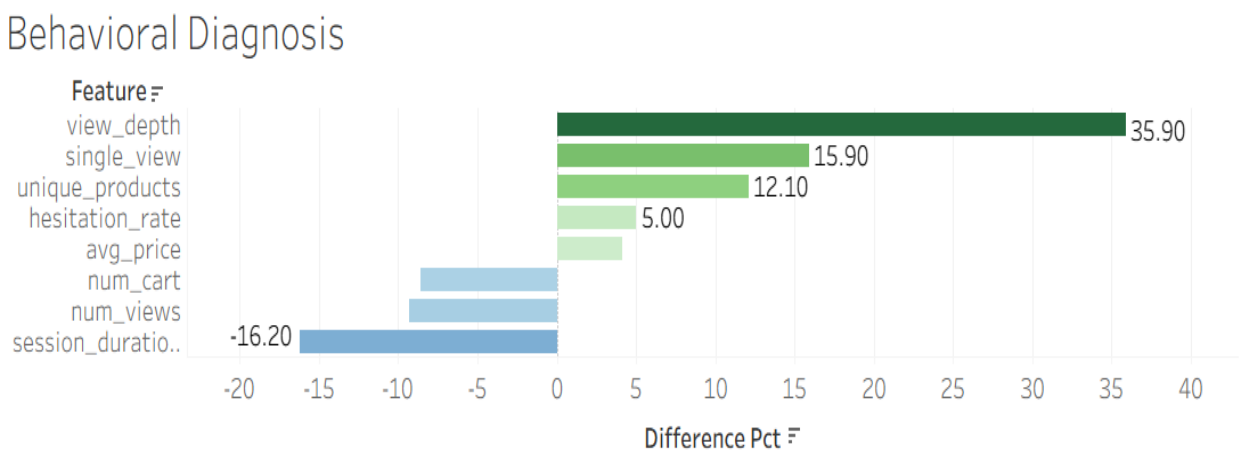

Figure 3: Dominant Inhibitors in High-Intent Non-Converting Sessions

Feature	Converted	Not Converted	Difference %
view_depth	4.10	3.02	+35.9%
total_events	27.59	20.59	+34.0%
single_view	0.167	0.144	+15.9%
unique_products	14.75	13.17	+12.1%
hesitation_rate	0.447	0.426	+5.0%
avg_price	7.41	7.12	+4.1%
max_price	19.50	19.25	+1.3%
num_cart	7.76	8.49	-8.6%
num_views	6.61	7.29	-9.3%
num_remove	4.53	4.82	-6.0%
session_duration_min	35.36	42.22	-16.2%

Several findings differ notably from the initial Logistic Regression-based analysis, reflecting the stronger discriminatory power of LightGBM segmentation.

The most striking reversal is session_duration_min, which is 16.2% lower in the converting group. Under LightGBM's more precise High Intent definition, converting users are goal-oriented and complete purchases quickly, without requiring extended browsing time. This contradicts the initial LR finding where duration was positive and suggests that time alone is not a reliable conversion signal within the High Intent group.

view_depth is now the strongest differentiator at +35.9%, indicating that converted users browse a wider variety of products per view event. This reflects focused, purposeful exploration rather than repetitive browsing of the same items. total_events remains a strong positive signal at +34.0%.

hesitation_rate has reversed from -3.2% (LR) to +5.0% (LGB), suggesting that within the High Intent group, converted users show slightly more cart removal behaviours. This may reflect active comparison shopping rather than indecision, as users add and remove items while evaluating alternatives before committing to a final purchase.

5.2 Dominant Inhibitors

Inhibitor	Evidence	Priority
Low browsing quality (view_depth)	view_depth 35.9% lower in non-converting group	High
Insufficient product exploration	unique_products 12.1% lower	High
Session duration	Non-converting sessions are longer (-16.2%), indicating aimless browsing	Medium
Price sensitivity	avg_price 4.1% higher in non-converting group	Medium
Hesitation	Slight positive difference (+5.0%); reflects comparison shopping, not a primary barrier	Low

6. Recommendations

6.1 Recommendation System

The recommendation strategy was designed based on the core diagnosis insight: converted High Intent users exhibit 35.9% higher view_depth, reflecting rational comparison behaviours across multiple products. Importantly, broader product exploration does not suppress spending — converted users have a 4.1% higher average price, indicating that product comparison builds purchase confidence rather than driving users toward cheaper alternatives.

The primary recommendation strategy is therefore to maximise product exposure depth, not to push cheaper substitutes. The system targets 309,658 High Intent sessions that added to cart but did not purchase, representing 99.2% of all High Intent non-converting sessions. This confirms that the Cart → Purchase gap is the dominant conversion problem for this audience.

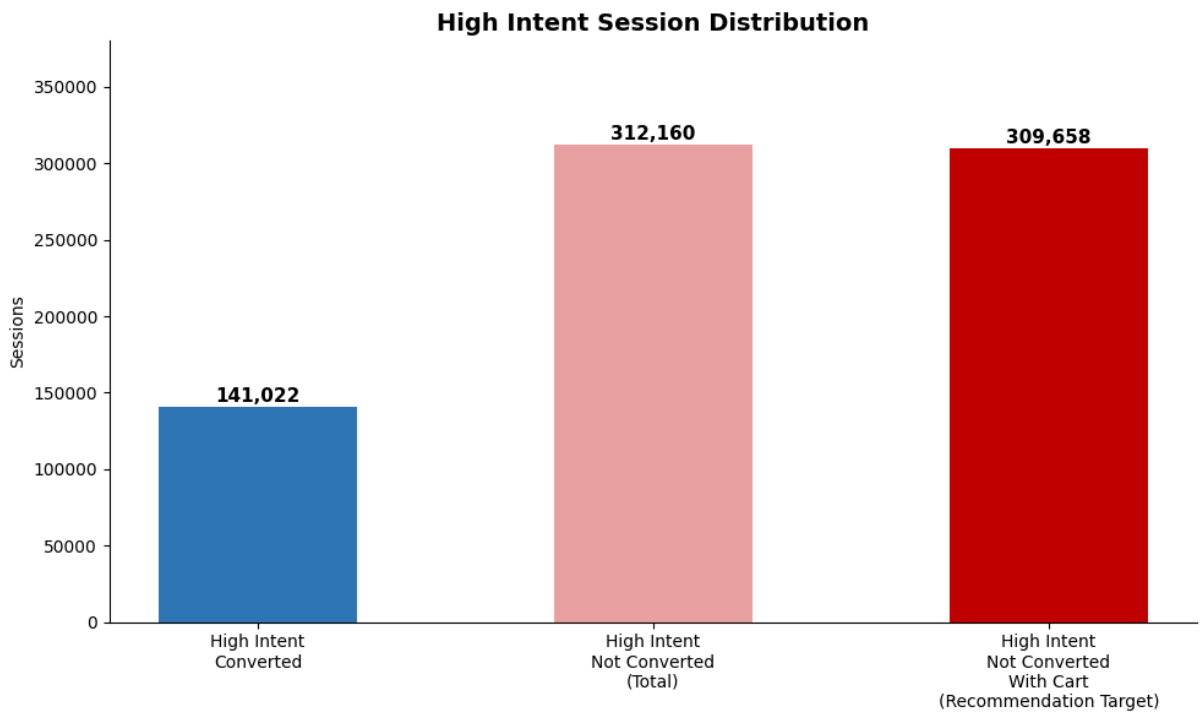

Figure 4: Recommendation System Architecture

Recommendation Logic:

For each target session, the system identifies the categories of cart items, retrieves products the user has not yet viewed within those categories, and ranks candidates by historical purchase volume.

Step	Logic
1. Identify cart categories	Extract category_id for all products added to cart in the session

Step	Logic
2. Find unseen products	Retrieve top-selling products in those categories not yet viewed in the session
3. Rank by popularity	Sort candidates by historical purchase_count (platform-wide)
4. Return top 3 per category	Surface at most 3 new products per category

The system was validated on a sample of 1,000 High Intent non-converting sessions, generating 9,683 recommendations with a 99.8% session coverage rate. Each session received an average of 9.7 product recommendations across its cart categories.

Exposure Tier Strategy:

Since the recommendation system operates through in-platform exposure rather than paid messaging, the marginal cost of an additional recommendation is near zero. Therefore, the trade-off shifts from monetary cost to traffic allocation efficiency. Sessions are ranked by predicted purchase probability and assigned to three exposure tiers:

Tier	Threshold	Strategy
High Exposure	Top 20% by probability	Priority placement, featured recommendation carousel
Normal Exposure	50th – 80th percentile	Standard recommendation module
No Exposure	Below 50th percentile	Excluded from recommendation engine

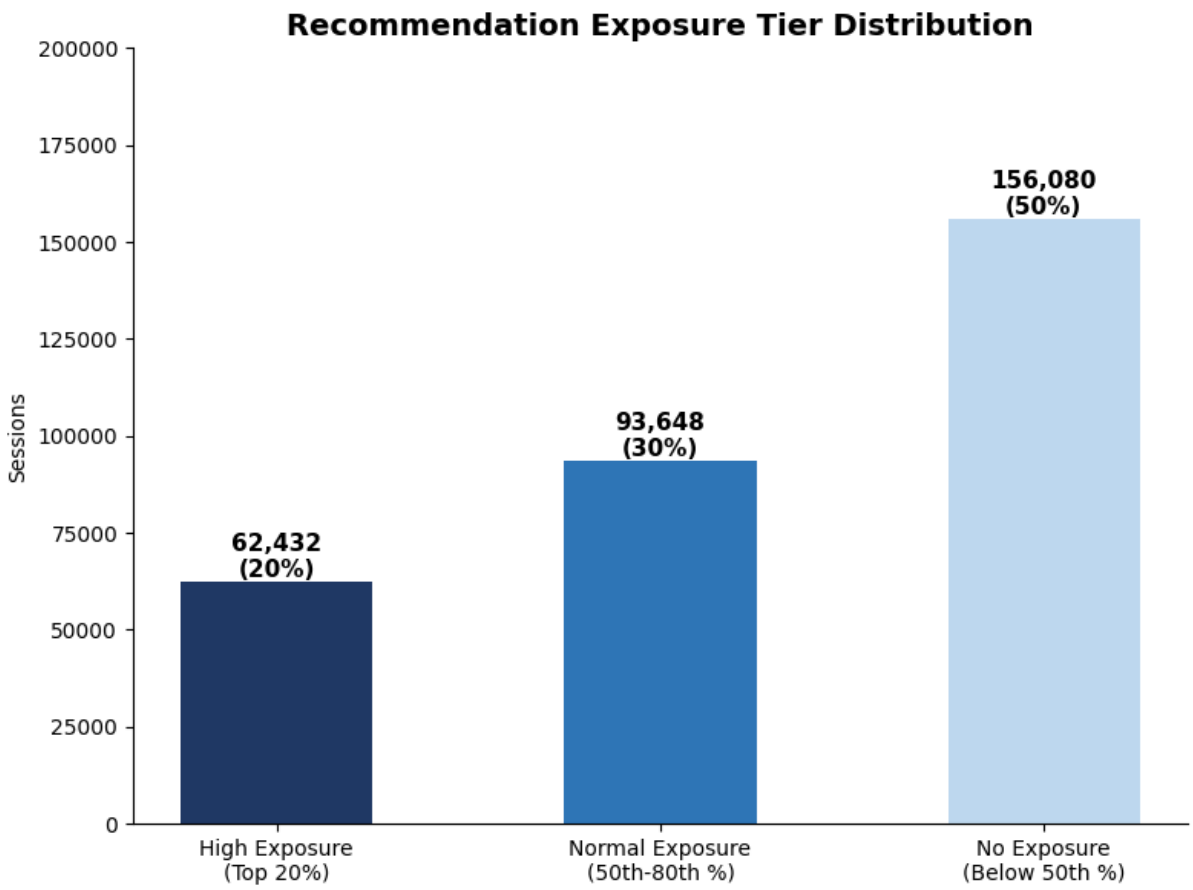

Figure 5: Exposure Tier Strategy Visualisation

Repeat-View Nudge:

For users who view the same product more than twice within a session without adding to cart, an in-page nudge is recommended with a direct purchase call-to-action. This targets high-interest users and shortens the decision path without requiring price reduction.

6.2 Strategic Recommendations

Finding	Recommendation	Expected Impact
view_depth 35.9% lower in non-converting group	Recommend unseen top-selling products in cart categories	Increase exploration depth and purchase confidence
Converted users pay 4.1% more	Push broader discovery, not cheaper alternatives	Protect and grow average order value
Non-converting sessions 16.2% longer	Surface goal-oriented product lists to shorten decision path	Reduce aimless browsing, increase conversion speed
Repeat product views without purchase	In-page nudge with direct purchase CTA after 2+ views	Convert high-interest sessions without price incentive

6.3 Setbacks and Limitations

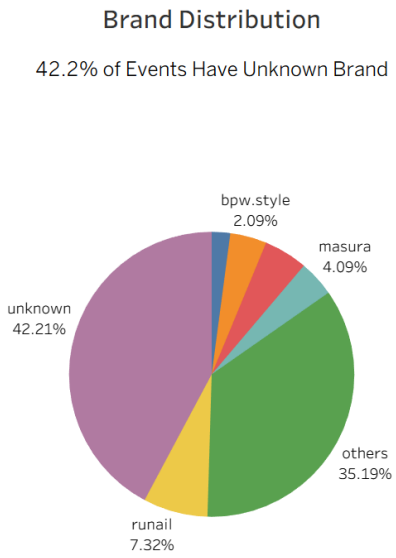

Figure 6: Brand Distribution Including Unknown Brand Breakdown

- Category analysis was severely limited by 98% missing category_code values. Recommendations operate on category_id numbers without human-readable labels, making it impossible to present category names to users without a lookup table.
- Brand data is incomplete, with 42.2% of all events (8.26 million records) recorded as unknown brand. This significantly limits brand-level analysis and prevents the recommendation system from incorporating brand preference signals, which are likely to be important in a professional beauty market where practitioners often demonstrate strong brand loyalty. Users who selected a premium brand may not respond positively to recommendations from different brands within the same category.
- The dataset covers only October 2019 to February 2020, limiting seasonal generalisability.
- No demographic, device, or traffic source data was available to enable persona-based personalisation beyond session-level behavioural signals.

6.4 Further Data Required

Data Required	Purpose
category_id to category_name lookup table	Enable human-readable category analysis and user-facing recommendations
User purchase history across sessions	Enable personalised recommendations beyond single-session signals

Data Required	Purpose
Traffic source and device type	Understand which channels produce highest-intent users
Post-session conversion data	Measure effectiveness of cart reminder and push campaigns

6.5 Next Steps

- Priority 1: Build category_id to category_name mapping table in collaboration with product team
- Priority 2: Add brand-aware filtering to recommendation logic to respect brand preferences
- Priority 2: Deploy Friday promotional push campaign for High Intent non-converting users
- Priority 3: Upgrade from rule-based to collaborative filtering recommendation model
- Priority 3: Build two-stage prediction model — Stage 1 predicts View to Cart, Stage 2 predicts Cart to Purchase
- Priority 4: A/B test recommendation placements to measure actual conversion lift and validate timing strategy effectiveness

Appendix: Deliverables Summary

Data Files (CSV)

File	Description
1 merged_all.csv	Raw merged dataset before cleaning
2 events_cleaned.csv	Cleaned event-level data (19,579,651 rows)
3 sessions_cleaned.csv	Session-level aggregated data (4,535,940 rows)
4 sessions_featured.csv	Session data with engineered features
5 sessions_modeled_lgb.csv	Session data with LightGBM predicted probabilities and segments
6 recommendations_sample.csv	Generated recommendations for 1,000 High Intent non-converting sessions
6.1 diagnosis_behavioral.csv	Behavioural comparison: converted vs non-converted High Intent sessions

File	Description
6.2 diagnosis_day.csv	Day-of-week conversion rate analysis
6.3 diagnosis_hour.csv	Hourly conversion rate analysis
7 tableau_data.csv	Combined session data with funnel stage for Tableau
7.1 tableau_overview.csv	Summary metrics for overview dashboard
7.2 tableau_duration.csv	Session duration distribution
7.3 tableau_events.csv	Event type distribution
7.4 tableau_price.csv	Price percentile summary
7.5 tableau_funnel.csv	Purchase funnel data
7.6 tableau_segment.csv	Session segmentation by intent
7.7 tableau_hourday.csv	Hourly and daily conversion heatmap data
7.8 tableau_diagnosis.csv	Behavioural diagnosis comparison data
7.9 tableau_brands_pie.csv	Brand distribution including unknown breakdown

Notebooks (Jupyter)

File	Description
1 cleaning & EDA.ipynb	Data cleaning, missing values, duplicates, price handling, time zone conversion
2 funnel analysis & feature engineering.ipynb	Purchase funnel construction and session-level feature engineering
3 model & segmentation.ipynb	LightGBM modelling, threshold optimisation, session segmentation
4 diagnosis.ipynb	High Intent session behavioural and temporal diagnosis
5 recommendation.ipynb	Ranking-based recommendation system with exposure tier strategy
6 visualisation.ipynb	Python charts for report and presentation

Tableau Workbook

Sheet	Description
1. Overview	Total unique customers and sessions

Sheet	Description
2. Session Duration	Distribution of session length across 7 bins (0-1 min to 2-24 hours)
3. Event Distribution	Session count by funnel stage: View Only, Cart, Purchase
4. Price Analysis	Price distribution by percentile (P10 to Max)
5. Purchase Funnel	Three-stage funnel with session counts and conversion rates
6. Hourly Conversion Heatmap	Conversion rate by hour and day of week
7. Session Segmentation	Session count and conversion rate by intent segment
8. Behavioural Diagnosis	Feature-level difference between converted and non-converted High Intent sessions
9. Brand Distribution	Brand breakdown including 42.2% unknown brand highlight